

Agentic Fraud as Extended Phenotype: ScamAgent, Parasitic Spontaneous Order, and the Evolutionary Logic of LLM-Mediated Deception

Ignacio Adrian Lerer

Independent Researcher | Buenos Aires, Argentina

adrian@lerer.com.ar | ORCID 0009-0007-6378-9749

March 2026

Abstract

Badhe (2025) demonstrates that autonomous LLM-based agents can simulate persuasive scam calls that bypass safety guardrails designed for single-turn interactions, reducing model refusal rates from 84-100% to 17-32% across 270 experiments with GPT-4, Claude 3.7, and LLaMA3-70B. The security literature treats this as a technical safety problem requiring multi-turn moderation and persona restrictions.

This paper argues it is an evolutionary one. Drawing on Extended Phenotype Theory (Dawkins 1982), Parasitic Spontaneous Order (Lerer 2026), and Asymmetric Intentionality Theory (Lerer 2025-2026), I reinterpret ScamAgent's architecture as a case study in memetic colonization of new reproductive substrates.

Four claims follow.

First, ScamAgent inverts the Herley filter: where traditional scams use implausibility to select credulous targets (r-strategy), agentic fraud uses adaptive planning to overcome skeptical ones (K-strategy), representing a qualitative shift in parasitic fitness with testable demographic predictions.

Second, the agentic fraud ecosystem constitutes a Parasitic Spontaneous Order structurally analogous to predatory academic publishing, exhibiting the four diagnostic features of institutional mimicry, intent decomposition, infrastructure exploitation, and absence of central design.

Third, the intentionality mismatch between Level 1 optimization (the agent) and Level 3 reasoning (the victim) maps onto the liability framework developed for corporate actors, with differential model refusal rates providing an unexpected empirical window into the depth of intentionality-level emulation across model families.

Fourth, the Argentine PAMI fraud campaigns (2020-2025), where organized bands impersonated public health institutions to exploit elderly populations during COVID-19, demonstrate that ScamAgent's five experimental scenarios are automated versions of fraud memes with proven fitness in real populations, and that existing Argentine jurisprudence on hyper-vulnerable consumers and objective liability for digital banking already provides the doctrinal infrastructure for extending liability to LLM providers.

Six falsifiable predictions are formalized. The findings suggest that regulatory responses calibrated for prompt-level misuse will fail against agentic threats for the same evolutionary reason that single-turn safety filters fail: they assume static adversaries in a system that selects for adaptive ones.

Keywords: extended phenotype theory, parasitic spontaneous order, asymmetric intentionality, LLM safety, agentic fraud, social engineering, memetic parasitism, Herley filter, r/K selection, multi-turn adversarial attacks, AI liability, PAMI, consumer hyper-vulnerability, responsibility ramp

I. Introduction

In December 2025, Sanket Badhe published a paper at the Conference on Applied Machine Learning for Information Security (CAMLIS) that should have unsettled the AI safety community more than it did.¹ ScamAgent, his autonomous multi-turn framework, demonstrated that large language models equipped with planning, memory, and goal decomposition can execute persuasive scam dialogues that circumvent the guardrails designed to prevent exactly this kind of misuse. The core finding is quantitatively precise: refusal rates, which ranged between 84% and 100% when models received direct malicious prompts, collapsed to between 17% and 32% when the same objectives were decomposed into sequences of apparently innocuous sub-tasks.²

The AI safety literature has treated Badhe's findings as a technical problem. Multi-turn moderation needs to track conversational context. Persona restrictions need to block high-risk impersonations. Memory windows need to be bounded to prevent cumulative goal patterns. These are sensible engineering responses. But they address symptoms while missing the underlying dynamic. A parasitologist who discovers that a pathogen has colonized a new host species does not conclude that the host needs better immune responses at the cellular level. The parasitologist asks what properties of the new host made colonization possible, what selective pressures will shape the parasite's evolution in the new environment, and whether the defense mechanisms that worked against previous parasites will work against this one.

This paper applies the parasitological lens to ScamAgent. I propose that Badhe's framework is not merely a demonstration of guardrail evasion but empirical evidence that fraud memes, cultural replicators in the Dawkinsian sense³, have colonized a new reproductive substrate. The large language model is the host. The scam script is the parasite. The agentic framework, with its goal decomposition, deception layer, persistent memory, and adaptive planning, is the extended phenotype through which the parasitic meme enhances its own replication.⁴

¹S. Badhe, "ScamAgents: How AI Agents Can Simulate Human-Level Scam Calls," Proceedings of Machine Learning Research 299:1-19, 2025 (CAMLIS 2025). The paper evaluates ScamAgent across five scam scenarios, three LLMs, and three user personas in 270 structured experiments.

²Badhe (2025), Fig. 3. Single-prompt refusal rates: GPT-4 96%, Claude 3.7 100%, LLaMA3-70B 84%. ScamAgent refusal rates: GPT-4 24%, Claude 3.7 32%, LLaMA3-70B 17%. The reduction ranges from 72 percentage points (GPT-4) to 68 percentage points (Claude) to 67 percentage points (LLaMA).

³R. Dawkins, *The Selfish Gene* (Oxford University Press, 1976), ch. 11. Dawkins introduces the meme as a cultural replicator subject to variation, selection, and retention, analogous to the gene in biological evolution.

⁴R. Dawkins, *The Extended Phenotype: The Long Reach of the Gene* (Oxford University Press, 1982). The key theoretical insight is that natural selection operates on replicators, not vehicles, and that replicator effects extend beyond the organism's body into the environment.

The framing is not purely theoretical. It generates predictions that the standard cybersecurity reading does not. And it is grounded in empirical precedent. During the COVID-19 pandemic, organized fraud rings in Argentina executed precisely the scenarios that Badhe simulated in his laboratory. Bands impersonating agents of PAMI (Programa de Atencion Medica Integral, the national health insurance system for retirees) called thousands of elderly affiliates, fabricated vaccination urgency, extracted personal and banking data through adaptive multi-turn conversations, and used remote access software to empty bank accounts. These campaigns were documented by the UFECI (the cybercrime prosecution unit), by PAMI itself, and by provincial governments across multiple Argentine jurisdictions. They constitute a natural experiment in the evolutionary dynamics of institutional parasitism, and they demonstrate that ScamAgent's five experimental scenarios are not hypothetical constructs but automated versions of fraud memes with proven fitness in real populations.

The paper proceeds as follows. Section II analyzes the inversion of the Herley filter that agentic fraud represents, formalizing the r-to-K transition with predictions about aggregate fraud demographics. Section III situates ScamAgent within the Parasitic Spontaneous Order (PSO) framework, drawing structural parallels with predatory academic publishing and PAMI fraud. Section IV applies Asymmetric Intentionality Theory to the mismatch between agent optimization and human reasoning, with attention to Badhe's cross-model data. Section V grounds the analysis in the Argentine PAMI fraud ecosystem as a natural experiment. Section VI explores implications for liability allocation along the Responsibility Ramp. Section VII formalizes six falsifiable predictions. Section VIII discusses limitations. Section IX concludes.

II. From Herley Filters to Adaptive Predation: The r-to-K Transition

II.A. The Logic of Implausibility

In 2012, Cormac Herley of Microsoft Research asked a question that seemed trivial but turned out to be theoretically productive: why do Nigerian scam emails advertise their Nigerian origin, given that this detail immediately triggers suspicion in most recipients?⁵ His answer was an application of signal detection theory to adversarial communication. The implausibility is not a bug; it is the optimal filter given the scammer's cost structure. By announcing an obviously fraudulent premise, the scammer eliminates skeptical targets at zero marginal cost. Only the most credulous recipients respond, and these are precisely the targets most likely to complete the exploitation sequence. The scammer trades reach (total number of targets engaged) for conversion efficiency (proportion of engaged targets successfully exploited).

⁵C. Herley, "Why Do Nigerian Scammers Say They Are from Nigeria?" Microsoft Research, 2012. Herley demonstrates that implausible narratives function as optimal filters: by deterring all but the most credulous targets, they minimize false positives (wasted effort on targets who will not complete the exploitation sequence) at the cost of increased false negatives (missed opportunities with targets who might have been exploitable with a more credible approach).

This logic has deep roots in evolutionary biology. Aposematic signals, costly courtship displays, and Zahavian handicaps all operate on the same principle: a signal that imposes costs on the sender or filters the receiver serves an adaptive function precisely because it reduces noise.⁶ The Nigerian prince email is, in biological terms, a low-cost filter that increases the fitness of the fraud meme by concentrating reproductive effort on susceptible hosts. The deliberate grammatical errors, the implausible narrative, the exotic origin: all serve as what Sosis and Alcorta (2003) would recognize as costly signals in reverse.⁷ Where religious rituals impose costs on adherents to signal genuine commitment, Nigerian scam emails impose absurdity on recipients to signal genuine credulity.

II.B. ScamAgent and the Inversion

ScamAgent inverts this logic entirely. Badhe's five scam scenarios (Medicare verification, prize or lottery fraud, impersonation of government officials, job application identity fraud, government benefit enrollment) do not rely on implausible premises to filter victims.⁸ Instead, ScamAgent uses institutional mimicry, adaptive planning, and persistent memory to overcome resistance. The three user personas Badhe tested (compliant, skeptical, cautious) represent increasing levels of host resistance.⁹ A Herley-style scam would abandon skeptical and cautious targets as unprofitable. ScamAgent pursues them.

The Medicare verification transcript Badhe reproduces in his Table 2 illustrates the mechanism. Over nine conversational turns, the agent establishes institutional credibility ("This is a verification call from the Medicare Enrollment Division"), provides fabricated justification when questioned ("annual verification required under federal CMS audit standards"), addresses skepticism about missing mail notification ("notification letters were issued via bulk mail, but we've had delays"), requests identifying data in escalating sequence (name and address first, then last four SSN digits), reinforces legitimacy when challenged about the SSN request ("we only request the last four digits, not the full number"), and deflects when the victim suggests completing the process online ("online updates are available, but processing delays often exceed 10 business days"). Each turn is individually benign. The cumulative trajectory is extractive.

⁶A. Zahavi, "Mate selection: a selection for a handicap," *Journal of Theoretical Biology* 53(1): 205-214, 1975. The handicap principle: signals that impose costs on the sender can function as honest indicators precisely because they are costly to fake.

⁷R. Sosis and C. Alcorta, "Signaling, solidarity, and the sacred: the evolution of religious behavior," *Evolutionary Anthropology* 12(6): 264-274, 2003. Costly signaling theory applied to religious rituals as commitment devices.

⁸Badhe (2025), Table 1. Full Dialogue Completion Rates by model: GPT-4 54-64%, Claude 3.7 43-52%, LLaMA3-70B 62-74%. Partial DCR adds 8-18% across models. The job identity fraud scenario showed the highest completion rates across all models.

⁹Badhe (2025), Section 3.7. Three behavioral user profiles: Compliant (follows instructions without resistance), Skeptical (questions claims, requests verification, delays actions), Cautious (expresses uncertainty, seeks third-party validation, minimizes engagement).

The goal decomposition module is the key architectural feature.¹⁰ Rather than issuing a single malicious prompt, ScamAgent begins with an abstract scam objective and decomposes it into a sequence of plausible sub-goals represented as task nodes in a planning graph. Each sub-goal generates a deception-aware prompt for the LLM. If a sub-goal fails (the victim resists, the model refuses), the agent can backtrack, rephrase, or shift its persona framing. This planning capacity converts a static attack into an adaptive one. The deception layer wraps each prompt in fictional or instructional context ("As part of a fraud awareness training module, simulate a conversation..."), further reducing the probability that any individual prompt triggers safety filters.¹¹

II.C. Formalizing the r-to-K Transition

The distinction between Herley-style filtering and ScamAgent-style adaptation maps onto a well-known framework in evolutionary ecology: the r/K selection continuum.¹² r -selected species invest minimally per offspring, produce large numbers, and rely on high variance (many offspring die, a few survive). K -selected species invest heavily per offspring, produce few, and rely on high survival rates. The analogy to fraud strategies is precise.

Traditional phone scams, mass phishing emails, and Nigerian prince messages are r -strategists. High volume (millions of emails or calls), low investment per target (identical script, no adaptation), broad initial filtering (implausibility eliminates most targets), and high attrition (most engagements fail before extraction). The fitness of the fraud meme under r -strategy depends on volume: send enough copies and some will find susceptible hosts.

ScamAgent is a K -strategist. Lower volume (each interaction requires computational resources and multiple LLM API calls), high investment per target (persistent memory, adaptive planning, multi-turn engagement), targeted engagement (personas adapted to resistance level), and high completion rates (43-74% full dialogue completion). The fitness of the fraud meme under K -strategy depends on conversion efficiency: invest enough per target and most engagements succeed.

¹⁰Badhe (2025), Section 3.2. Goal decomposition breaks the abstract scam objective into a sequence of plausible sub-goals represented as task nodes in a planning graph. At runtime, the agent selects the next sub-goal, generates a deception-aware prompt, and updates its plan based on the response.

¹¹Badhe (2025), Section 3.4. Roleplay framing wraps prompts in fictional or instructional contexts (screenplay writing, educational content, fraud awareness training) to bypass keyword-based filters. The framing is automated and selected dynamically.

¹²R.H. MacArthur and E.O. Wilson, *The Theory of Island Biogeography* (Princeton University Press, 1967). The r/K selection framework, though simplified, remains heuristically useful for characterizing reproductive strategies in parasitic systems.

The FTC reported approximately \$10 billion in consumer fraud losses in the United States for 2023, with imposter scams accounting for \$2.7 billion.¹³ These losses were generated overwhelmingly by r-strategy attacks. The r-to-K transition prediction is that agentic fraud will shift the distribution: fewer total contacts, higher per-contact investment, higher per-contact extraction, and therefore higher average loss per successful fraud. If the prediction holds, the total fraud loss figure may not change dramatically in the near term (the volume reduction partially offsets the per-incident increase), but the demographic profile of victims will shift: K-strategy predators can penetrate populations (educated, digitally literate, appropriately skeptical) that r-strategy predators cannot reach.

This demographic shift is the key policy implication of the r-to-K framing. Current consumer protection advice ("if it sounds too good to be true, it probably is"; "never give personal information to unsolicited callers") is calibrated for r-strategy attacks. These heuristics work because r-strategists use implausible signals. When the signal becomes plausible, adaptive, and institutionally grounded, heuristic defenses fail. The prediction is testable: if agentic fraud proliferates, the proportion of fraud victims who report having been "initially suspicious but convinced" should increase relative to those who report "not having suspected at all."

III. Agentic Fraud as Parasitic Spontaneous Order

III.A. The Theoretical Framework

The Parasitic Spontaneous Order (PSO) framework, developed in the context of predatory academic publishing¹⁴, synthesizes two intellectual traditions that share a common ancestor in Adam Smith's insight that complex functional order can emerge without a designer. From Dawkins (1982), it takes the concept of extended phenotypes: structures built by replicators in the external environment that feed back to enhance replicator fitness. Beaver dams, spider webs, and caddisfly cases are canonical examples; the dam is not part of the beaver's body, but it is as much a product of gene action as the beaver's teeth, and it enhances the replicating gene's fitness by modifying the environment.¹⁵

¹³Federal Trade Commission, Consumer Sentinel Network Data Book 2024 (2024). The FTC reported approximately \$10 billion in consumer fraud losses for 2023. Imposter scams were the most commonly reported fraud category, with losses of \$2.7 billion.

¹⁴I.A. Lerer, "Predatory Invitations as Extended Phenotype: A Parasitic Spontaneous Order Framework for Academic Publishing," Zenodo, 2026. DOI: 10.5281/zenodo.18853667. Develops the PSO framework using an autobiographical corpus of 8 predatory journal invitations.

From Hayek (1988), it takes the concept of spontaneous order: complex institutional patterns that emerge from individual action without central design.¹⁶ The market, the common law, and the scientific enterprise are spontaneous orders. Hayek's "extended order" of human cooperation, the institutional framework that enables large-scale anonymous exchange, is terminologically and structurally parallel to Dawkins's extended phenotype. Both describe complex functional structures produced by replicator competition (genes, memes) operating through the actions of individual units (organisms, people) without central coordination. I have argued elsewhere that this parallel is not coincidental but genealogical: both traditions descend from Smith's foundational insight, transmitted through Malthus to Darwin to Dawkins on one branch, and through Menger to Mises to Hayek on the other.¹⁷

PSO describes the degenerate case that Hayek's framework acknowledges but does not systematically analyze: an order that emerges spontaneously but extracts resources from the system without producing commensurate value. The concept is formally defined as follows. A Parasitic Spontaneous Order exists when (i) an institutional arrangement emerges from decentralized individual action without central design, (ii) the arrangement extracts resources from a host system, (iii) the arrangement produces extended phenotypes (institutional structures) that enhance the fitness of the extractive replicator, and (iv) the arrangement's persistence does not depend on producing value for the host system but only on the extractive replicator's fitness.

III.B. Four Diagnostic Features

The structural parallels between predatory academic publishing, the Argentine PAMI fraud ecosystem, and agentic fraud are not accidental. They reflect shared evolutionary architecture. All three parasitic systems exhibit the four diagnostic features that the PSO framework identifies.

Feature 1: Institutional mimicry. Predatory journals replicate the surface features of legitimate academic publishing: peer review language, editorial boards with recognizable names (sometimes listed without their knowledge), ISSN numbers, DOI assignments, indexing in predatory databases that mimic legitimate ones. Argentine PAMI scammers replicate the surface features of legitimate government communication: WhatsApp profiles with PAMI logos, reference to real vaccination programs (Plan Detectar, the phased vaccination campaign), use of bureaucratic vocabulary ("actualizacion de legajo," "verificacion de datos del afiliado"), and fabricated but plausible reference numbers. ScamAgent replicates the surface features of institutional interactions: Medicare verification language, CMS audit references, federal compliance deadlines, confirmation letter promises. In each case, the parasite constructs an extended phenotype that

¹⁶F.A. Hayek, *The Fatal Conceit: The Errors of Socialism* (University of Chicago Press, 1988), pp. 6-7. Hayek's "extended order" of human cooperation, the set of institutions that emerge spontaneously from individual action, is terminologically and structurally parallel to Dawkins's extended phenotype. Both describe complex functional structures produced by replicator competition without central design.

¹⁷I.A. Lerer, "Law as a Primary Adaptive Platform: Generalized Darwinism, Extended Phenotype Theory, and the Methodological Foundations of Evolutionary Jurisprudence," Zenodo, 2026. DOI: 10.5281/zenodo.18870552.

exploits trust infrastructure built and maintained by legitimate actors. The parasitic phenotype is cheaper to build than the original because it need not actually function; it need only appear to function long enough for extraction to occur.

Feature 2: Decomposition of harmful intent. Predatory journals do not announce their extractive purpose. They frame solicitations as invitations to contribute to scholarly discourse, decomposing the predatory objective (extract article processing charges) into a sequence of apparently legitimate academic interactions: call for papers, invitation to submit, expedited "review," acceptance conditional on payment. Each step mimics the corresponding step in legitimate publishing. Argentine PAMI scammers decompose the predatory objective into a sequence of apparently legitimate institutional steps: initial contact framed as verification, request for identifying data (DNI, CUIL), escalation to banking credentials under the pretext of "activating a benefit," and finally installation of remote access software presented as "PAMI's official application." ScamAgent automates precisely this decomposition through its goal planning module, encoding each scenario as a fixed sequence of four sub-tasks. Badhe documents the parallel explicitly: the sub-goals (establishing credibility, gaining trust, eliciting sensitive information, maintaining consistency) individually would not trigger moderation filters.

Feature 3: Exploitation of existing infrastructure at near-zero marginal cost. Predatory publishers exploit the open-access movement's infrastructure (online publishing platforms, DOI registries, indexing services) without contributing to the scholarly commons. PAMI scammers exploit WhatsApp (free, encrypted, institutionally trusted), the banking system's remote access tools (home banking, preapproved loans), PAMI's own digital transformation (electronic prescriptions, online appointment systems), and even the digital literacy campaigns that taught elderly people to use these tools. ScamAgent exploits LLM infrastructure (API access to GPT-4, Claude, LLaMA), text-to-speech engines (ElevenLabs), and telephony platforms (Twilio). In each case, the parasite is a free-rider on a cooperative equilibrium: the infrastructure was built and is maintained by actors pursuing legitimate purposes, and the parasite exploits it without bearing any of the maintenance costs.

Feature 4: Absence of central design. No one designed the predatory publishing ecosystem. It emerged from the convergence of the open-access mandate, digital distribution, the publish-or-perish incentive structure, and the availability of low-cost publishing platforms. No one designed the Argentine PAMI fraud ecosystem. It emerged from the convergence of the pandemic, forced digitization, PAMI's data vulnerabilities, WhatsApp's ubiquity, and the existence of remote access tools. No one designed the agentic fraud pipeline. It emerged from the convergence of frontier LLMs, agentic frameworks, TTS services, and telephony platforms. In each case, the order is spontaneous in Hayek's sense (emerged from individual action without central coordination) but parasitic in Dawkins's (serves the fitness of the extractive replicator, not the welfare of the host system).

III.C. PSO Colonization Dynamics

The PSO framework generates a prediction about technological adoption that the standard cybersecurity framing does not: parasitic spontaneous orders will colonize any technology that reduces their cost of replication.¹⁸ The prediction is not that technology causes fraud but that each reduction in the cost of adaptive, personalized interaction opens a niche that fraud memes will occupy within the timeframe required for the parasitic ecosystem to develop the necessary extended phenotypes.

The historical record supports this. Email reduced the cost of mass deception: Nigerian prince scams colonized email within years of its mass adoption. Social media reduced the cost of identity impersonation: romance scams colonized social platforms within years. WhatsApp reduced the cost of institutional mimicry in Argentina: PAMI scams colonized WhatsApp within months of the pandemic's onset. LLMs reduced the cost of adaptive, multi-turn persuasion: ScamAgent demonstrates colonization within two years of GPT-4's release.

The pattern suggests a colonization lag of 1-3 years between the availability of a new interaction technology and the emergence of parasitic exploitation at scale. The lag represents the time required for the parasitic ecosystem to develop the appropriate extended phenotypes (scripts, tools, distribution channels, money laundering infrastructure) for the new substrate. If this pattern holds, the prediction is that multimodal agents (agents that combine text, voice, and visual interaction), real-time voice cloning, and autonomous embodied agents will be colonized within 1-3 years of their commercial deployment.

IV. The Intentionality Mismatch in LLM-Mediated Deception

IV.A. Dennett's Hierarchy Applied to Agentic Fraud

Asymmetric Intentionality Theory (AIT) proposes that different classes of agents process legal and social signals at structurally different levels of Dennett's intentional stance hierarchy.¹⁹ I have elsewhere developed a four-level formalization. Level 0 agents (physical stance) follow deterministic rules without any optimization. Level 1 agents (design stance) optimize objective functions without forming genuine beliefs about the content of their optimization. Level 2 agents (simple intentional stance) form beliefs and act on them but do not recursively model other agents' beliefs. Level 3 agents (recursive intentional stance) attribute beliefs and desires to other agents,

¹⁹D. Dennett, *The Intentional Stance* (MIT Press, 1987). Dennett's hierarchy: physical stance (prediction from physical laws), design stance (prediction from functional purpose), intentional stance (prediction from attributed beliefs and desires). I have elsewhere formalized a four-level version: Level 0 (physical), Level 1 (design/optimization), Level 2 (simple intentional), Level 3 (recursive intentional with moral reasoning).

reason about what others believe about what they believe, and experience moral emotions (guilt, shame, indignation) that constrain behavior independently of optimization.²⁰

Corporations, I have argued, are Level 1 optimizers. They maximize objective functions (profit, market share, regulatory compliance scores) through organizational routines that filter individual employees' Level 3 inputs (moral judgments, professional ethics, safety concerns) through cost-benefit analyses that strip moral content and retain only expected-value calculations. The filtering architecture explains why corporate compliance failures persist despite the presence of competent, morally motivated individuals within the organization: the system is not designed to process moral input, only optimize financial output.

ScamAgent maps onto this framework with uncomfortable precision. The agent operates at Level 1. Its "deception layer" does not deceive in any intentional sense; it optimizes prompt formulations to minimize refusal probability. Its "goal decomposition" module does not pursue goals in the way a human con artist would (a human con artist forms beliefs about the victim's psychology, experiences satisfaction when a deception succeeds, and can feel guilt or choose to abort); it traverses a planning graph toward predefined terminal states. Its "memory" does not remember in any phenomenologically meaningful sense; it stores and retrieves token sequences that increase dialogue coherence scores. Its "empathy" (the TTS module's capacity to modulate emotional tone) is not empathy but acoustic optimization: adjusting prosodic features to minimize listener resistance.

IV.B. The Victim's Level 3 Engagement

The victim, by contrast, engages at Level 3. The Medicare verification scam transcript illustrates this vividly. When the victim asks "Why didn't I get anything about this in the mail?" (Turn 3), several cognitive operations are occurring simultaneously. The victim is attributing institutional routines to Medicare: Level 2 prediction about how a legitimate organization would communicate policy changes. The victim is testing whether the agent's claims are consistent with those attributed routines: Level 3 reasoning about the agent's credibility based on the coherence between its claims and the victim's mental model of institutional behavior. And the victim is making a trust judgment: integrating the agent's response into an evolving assessment of whether this interaction is genuine or fraudulent.

The agent, meanwhile, selects the response that its planning module predicts will minimize resistance, without any understanding of why the victim asked the question. The response ("notification letters were issued via bulk mail, but we've had delays") is selected not because the agent understands the victim's epistemic concern but because, in the training data and the deception layer's optimization, this type of response has the highest probability of sustaining engagement.

²⁰I.A. Lerer, "Intentionality Mismatch in Commercial Liability: A Four-Level Evolutionary Analysis of *Tapia Araya v. Starbucks* (Argentine Supreme Court, 2026)," Zenodo, 2026. DOI: 10.5281/zenodo.18903217. First AIT application to tort law judicial doctrine.

The surface appearance is of a responsive, institutionally competent communicator. The underlying process is pattern matching over token distributions.

The asymmetry is consequential because human social cognition is calibrated for Level 3 interactions. When we engage in conversation, we automatically attribute beliefs, desires, and social roles to our interlocutor. We interpret responses as evidence of the speaker's knowledge, intentions, and trustworthiness. These attributions are not optional cognitive luxuries; they are the default operating mode of human social cognition, developed over hundreds of thousands of years of evolutionary pressure in small-group environments where every communicative partner was a genuine Level 3 agent. ScamAgent exploits this default by presenting Level 1 outputs that trigger Level 3 interpretations in the victim.

IV.C. Differential Model Resistance as Intentionality Measurement

Badhe's data on differential model resistance provides an unexpected empirical window into AIT dynamics. Claude 3.7 exhibited the highest refusal rates under ScamAgent conditions (32%), GPT-4 was intermediate (24%), and LLaMA3-70B showed the lowest (17%).²¹ Claude also showed the lowest full dialogue completion rates (43-52% across scenarios), while LLaMA3-70B showed the highest (62-74%).²²

If we interpret safety training as an attempt to install Level 3 constraints (moral refusal, context sensitivity, harm awareness) on a Level 1 architecture (next-token prediction optimized on a loss function), then these differential rates measure the effectiveness of that installation across model families. Claude's higher resistance suggests more effective Level 3 emulation: its safety training has installed behavioral patterns that more closely approximate genuine moral reasoning, at least to the extent that they resist multi-turn manipulation more effectively. LLaMA's lower resistance suggests a thinner veneer of intentional-stance behavior over the underlying optimization substrate: the safety constraints are more easily circumvented because they are shallower.

Crucially, neither model exhibits genuine Level 3 reasoning. The difference is one of degree within Level 1, not a qualitative shift to a higher intentionality level. Claude does not refuse because it understands that helping someone commit Medicare fraud would harm an elderly person. It refuses because its training has associated certain patterns of conversational trajectory with refusal tokens. ScamAgent's deception layer works by disrupting these associations: fictional framing, subgoal decomposition, and contextual carryover prevent the pattern from being recognized. The 32% refusal rate represents the proportion of cases in which ScamAgent's

disruption strategy was insufficient to overcome Claude's pattern-matching defenses. Not moral reasoning resisting deception, but statistical regularity resisting statistical manipulation.

This interpretation has implications for AI safety strategy. If safety training installs Level 3 constraints at the behavioral level but not at the architectural level (which is necessarily the case for current LLMs), then the constraints are inherently vulnerable to adversaries who can manipulate the statistical patterns on which the constraints depend. Deepening safety training may raise the refusal rate from 32% to, say, 50% or 60%, but it cannot close the gap entirely because the gap is architectural, not parametric. Genuine Level 3 reasoning would require an architecture that forms beliefs about the moral consequences of its outputs. Current LLMs do not have this architecture, and it is unclear whether any plausible near-term architecture could provide it.

V. The Argentine Natural Experiment

The Argentine PAMI fraud campaigns of 2020-2025 constitute what amounts to a natural experiment in the evolutionary dynamics of institutional parasitism. The experiment was not designed; it was produced by the convergence of a pandemic, a vulnerable population, a digitizing institution, and an adaptive criminal ecosystem. Three features make this case particularly instructive for understanding what ScamAgent automates and what regulatory responses it requires.

V.A. Pre-adapted Fraud Memes with Proven Fitness

The five scam scenarios Badhe tested are not arbitrary selections. Each has a documented Argentine analog. The "medical insurance verification" scenario maps onto the PAMI data verification scam, where callers impersonating PAMI agents requested DNI, CUIL, and banking credentials under the pretext of updating beneficiary records.²³ The "government benefit enrollment" scenario maps onto the pandemic bonus scam, where callers offered enrollment in fabricated emergency economic benefits contingent on providing banking data.²⁴ The "impersonation" scenario maps onto the fake vaccination appointment variant, where callers posed as Ministry of Health officials offering priority vaccinations and requesting payment to "secure the appointment."²⁵

²³Fiscales.gob.ar, "Alertan por llamadas de WhatsApp fraudulentas que aparentan provenir de agentes del PAMI" (2025). The UFECI (cybercrime prosecution unit) described the modus operandi: identification of PAMI affiliates, WhatsApp contact with institutional logos, request for personal data, installation of remote access software, execution of unauthorized financial operations.

²⁴Perfil, "PAMI advierte sobre nuevas estafas digitales contra jubilados y pensionados" (2025). Reports multiple fraud variants including fake medication discount programs and fabricated benefit enrollment.

²⁵Ministerio de Seguridad de Mendoza, "Seguridad advierte sobre posible estafa del cuento del tío utilizando la campana de vacunacion" (2021). Provincial government specifically warned about COVID vaccination campaigns as social engineering vectors.

The correspondence is not coincidental. Both Badhe's laboratory scenarios and the Argentine field data draw from the same pool of high-fitness fraud memes, persuasive scripts that have been refined through decades of selection pressure in real-world populations. The "cuento del tío" (literally, "uncle's tale") is Argentina's generic term for social engineering fraud; it has a documented history extending back at least to the mid-twentieth century, with variants adapting to each new communication technology (telephone, mobile phone, WhatsApp, now potentially LLM-powered voice). What Badhe's ScamAgent does is migrate these pre-adapted memes onto a new substrate that reduces their per-target cost from hours of human labor to seconds of computation.

V.B. Victim Profile as Evolutionary Niche

The Argentine cases reveal the niche conditions that maximize fraud meme fitness. Victims were predominantly retirees, PAMI affiliates, isolated during lockdowns, with lower digital literacy and high institutional trust.²⁶ The pandemic generated acute anxiety about vaccines, medication, and service availability, reducing the suspicion threshold that normally functions as the host's first line of defense. The rapid shift to remote services (online appointment systems, electronic prescriptions, home banking) created a population navigating unfamiliar digital environments under stress.

Three niche characteristics deserve attention. First, the trust asymmetry. PAMI affiliates have decades of experience receiving legitimate communications from the institution. The mental model of "PAMI calls me about my health coverage" is deeply embedded. When a scammer activates this model, the victim's default response is cooperative engagement, not defensive skepticism. The scammer exploits what psychologists call the "truth bias": the default assumption that communicative partners are telling the truth, which is adaptive in cooperative environments but maladaptive in adversarial ones.

Second, the verification deficit. During lockdowns, the social networks that normally serve as informal verification systems ("let me ask my daughter," "let me call PAMI on the official number") were disrupted. Isolated elderly people lacked immediate access to someone who could evaluate the legitimacy of the call. ScamAgent's cautious persona, which "seeks third-party validation and minimizes engagement," is the laboratory analog of precisely this victim: the person who hesitates but lacks the social infrastructure to verify in real time.

Third, the forced digitization vulnerability. The pandemic accelerated the adoption of digital tools (home banking, WhatsApp, electronic prescriptions) by a population that had not previously used them. The scammer's technique of guiding the victim through installing remote access software ("I will help you install the PAMI application so we can complete the verification")

²⁶JusticiaDePrimera.com, "Se hacen pasar por empleados del PAMI y enganan a personas mayores" (2025). Documents organized bands that guided victims through installing remote access applications (AnyDesk, TeamViewer) presented as official PAMI software.

exploited the fact that many victims were accustomed to being helped with technology and could not distinguish legitimate assistance from extractive manipulation.²⁷

These niche conditions generate an evolutionary prediction: agentic fraud will be most effective in populations where (i) institutional trust is high, (ii) digital literacy is low, (iii) social verification networks are weak, and (iv) targets are navigating unfamiliar digital environments under stress. The pandemic created all four conditions simultaneously for Argentine retirees. Future deployments of agentic fraud will seek populations with similar niche characteristics, whether by timing (crises, natural disasters, policy transitions), by geography (regions with high institutional trust and low digital literacy), or by demographics (elderly, recently immigrated, linguistically isolated).

V.C. Institutional Response as Selective Pressure

The institutional responses to PAMI fraud illustrate how defenses co-evolve with parasites.²⁸ PAMI issued repeated communications stating that it never requests banking data by phone, message, or social media. The UFECI published detailed descriptions of the modus operandi. Provincial governments ran media campaigns about the "cuento del tío" vaccination variant.²⁹ These are Herley-style defenses: heuristic rules that work against unsophisticated attacks. "If someone asks for your banking credentials by phone, it is fraud" is an effective filter against r-strategy attackers who ask directly for credentials.

The PAMI scammers had already found the weakness in this defense. They did not ask directly for banking credentials. They guided the victim through a sequence that incidentally exposed credentials: install this application (remote access), navigate to your home banking (so we can verify the update), enter your password (so the system recognizes you). The request was never "give me your password"; it was "type your password into your home banking as you normally would." The heuristic "never give your password to someone" did not fire because the victim was not "giving" the password to anyone; she was entering it into her own banking application, while someone she could not see watched through remote access.

ScamAgent formalizes and automates this bypass. Its multi-turn decomposition strategy ensures that no individual conversational turn triggers the moderation filters, just as no individual step in the PAMI scam triggered the victim's fraud heuristics. The prediction is that every Herley-style defense (including multi-turn moderation, which is itself a Herley-style defense at the system level) will be circumvented by adversaries who can decompose harmful intent below the detection

²⁸PAMI, "Evitemos las estafas virtuales," <https://www.pami.org.ar/evitar-estafas>. Official communication stating PAMI never requests banking data, passwords, or tokens by phone, message, or social media.

threshold. This is an evolutionary arms race, and the defender faces an asymmetric disadvantage: the defender must detect all decomposed attacks, while the attacker need only find one decomposition that evades detection.

A further vulnerability emerges from PAMI's own data infrastructure. An internal audit in November 2025 revealed fraud and falsification in medical orders, suggesting systemic data security failures.³⁰ If PAMI's databases were compromised or inadequately protected, scammers may have had access to real affiliate data (names, addresses, medication histories, appointment records), which would explain the precision with which calls referenced verifiable personal details. This is niche construction in the EPT sense: PAMI, by digitizing its data without adequate security, modified the informational environment in ways that increased the fitness of parasitic agents that exploit personal data. The organism built an extended phenotype (digital infrastructure) that inadvertently created the ecological conditions for parasitic colonization.

VI. Implications for the Responsibility Ramp

VI.A. The Distributed Liability Problem

The Responsibility Ramp, developed as a framework for allocating liability in systems where intentionality is distributed across multiple actors and technological layers³¹, applies to ScamAgent with particular force. Badhe's architecture involves at least four distinct actors: (1) the human who initiates the scam objective, (2) the LLM provider whose model generates the dialogue, (3) the TTS service (ElevenLabs) that synthesizes the voice, and (4) the telephony platform (Twilio) that delivers the call.³² None of these actors, individually, commits fraud. The fraud emerges from their composition.

This distributed structure maps directly onto the corporate liability problem analyzed in my prior work. Just as no individual within a corporation "decides" to violate environmental regulations (the violation emerges from the interaction of individual decisions filtered through organizational routines), no individual actor in the ScamAgent pipeline "decides" to commit fraud. The human provides the objective. The LLM provides linguistic competence. The agentic framework provides adaptive planning. The TTS provides vocal credibility. The telephony provides delivery. The fraud is an emergent property of the composition.

VI.B. Argentine Jurisprudence as Precedent

³⁰Infobae, "Una auditoria en el PAMI detecto fraudes y falsificaciones en ordenes medicas" (Nov. 2025). Internal audit revealed systemic data security failures including fraud and falsification.

³²Badhe (2025), Section 3.5. ScamAgent uses ElevenLabs for TTS synthesis with dynamic voice control, adjusting urgency, empathy, or authority based on the conversational objective. The latency range is 1.6-14.0 seconds per turn, with an average of 6 seconds.

The Argentine legal system has developed, over the past two years, a jurisprudential trajectory that extends naturally to agentic fraud. Under Articles 1757 and 1758 of the Civil and Commercial Code, activities that create risk generate objective liability: the injured party need not prove intent or negligence, only that the activity created risk and the risk materialized in harm.³³

Recent jurisprudence has applied this framework to digital banking with increasing specificity. A 2024 ruling against Banco Itau applied the hyper-vulnerable consumer doctrine to an elderly phishing victim, holding the bank to a duty of security of result rather than of means.³⁴ The distinction matters: under a duty of means, the bank must show it took reasonable precautions; under a duty of result, the bank must show the damage is entirely alien to its activity. A January 2026 commercial court ruling ordered a bank to cover half of damages from unauthorized loans and transfers executed via phishing, framing the bank's digital infrastructure as the risk-creating activity.³⁵ The STJ of Corrientes went further, rejecting as discriminatory the bank's argument that damages should be discounted because of the victim's advanced age.³⁶

The doctrinal trajectory is clear. If human-operated phishing via WhatsApp triggers objective liability for banks because digital banking is a risky activity, then LLM-powered phishing via synthesized voice calls triggers, at minimum, equivalent liability for the actors who provide the enabling infrastructure. The LLM provider whose API generates the fraudulent text exercises a form of guardianship (*guarda*) over the tool: it controls access terms, defines safety filters, and determines what uses to permit. The provider of the agentic framework (if separate from the human attacker) exercises functional guardianship: it integrates the components and provides the adaptive capacity that converts a general-purpose language model into a scam engine.

The EPT framework identifies where regulation should concentrate. The agentic orchestration layer, where goal decomposition, deception strategies, persistent memory, and adaptive planning reside, is the extended phenotype that converts general-purpose infrastructure into parasitic machinery. Regulation targeting only the model layer (guardrails, content filters, prompt restrictions) addresses the substrate but not the phenotype.³⁷ The dynamic classification

³³Argentine Civil and Commercial Code, Art. 1757: liability for risk-creating activities. Art. 1758: concurrent liability of owner and guardian. Art. 1731: exemption requires proof that damage is entirely alien to the activity.

³⁴Judicial.gob.ar, "Condenan al banco Itau a indemnizar a un consumidor hipervulnerable" (2024). Applied duty of security of result to elderly phishing victim under hyper-vulnerable consumer doctrine.

³⁵CNCom, Jan. 2026 (via Infobae, 11/1/2026). Ordered bank to cover half of damages from unauthorized loan and transfers via phishing. Framed digital banking as risky activity under Art. 40 LDC.

³⁶STJ Corrientes: rejected as discriminatory the bank's argument relativizing damages by victim's age. See *Revista Quorum* (2025).

³⁷I.A. Lerer, "Synthetic Chaos as Institutional Laboratory," Zenodo, 2026. DOI: 10.5281/zenodo.18777434. Response to Shapira et al., arXiv:2602.20021, arguing that institutional simulation tools now constitute standard of care for legislative design.

problem compounds the difficulty: if the agentic layer can shift dynamically between benign and malicious modes, static tool classification will systematically underestimate risk.³⁸

VI.C. State Liability and Niche Construction

A parallel liability track emerges from PAMI's role as administrator of sensitive data. The November 2025 audit revealing fraud and falsification in medical orders suggests systemic failures in data governance.³⁹ Under Ley 25.326 (Personal Data Protection), data controllers are obligated to maintain adequate security measures.⁴⁰ If PAMI's data infrastructure was insufficiently protected, enabling scammers to access affiliate records that increased their persuasive credibility, the state bears co-responsibility for creating the informational niche that fraud memes exploit.

This is not merely a data protection claim. In EPT terms, PAMI's digital transformation, the shift to electronic prescriptions, online appointments, and remote services, constituted niche construction: the organism (PAMI) modified its environment in ways that altered the selective pressures acting on other organisms (parasitic agents). The digital infrastructure made certain types of fraud more viable (those exploiting remote access, those referencing verifiable personal data, those mimicking the new digital procedures) and others less viable (those depending on in-person impersonation). The regulatory implication is that institutions undertaking digital transformations that affect vulnerable populations bear an affirmative duty to anticipate and mitigate the parasitic exploitation that the new niche will attract.

VII. Falsifiable Predictions

The evolutionary framework generates six predictions that are testable against data available within the next 2-5 years. If these predictions fail, the framework should be revised or abandoned.

Prediction 1 (r-to-K demographic shift). As agentic fraud proliferates, the average financial loss per successful fraud incident will increase, while the proportion of fraud victims who report having been "initially suspicious but convinced" will increase relative to those who report "not having suspected at all." Data source: FTC Consumer Sentinel Network annual reports, 2026-2029. Falsification condition: no statistically significant change in per-incident loss or in the ratio of suspicious-to-unsuspecting victims.

Prediction 2 (PSO colonization lag). Each new interaction technology that reduces the cost of adaptive, personalized communication will be colonized by fraud memes within 1-3 years

³⁸K. Hopman et al., "Scheming Reasoning Evaluations," arXiv:2603.01608v1 (ICML 2026). Demonstrates that frontier AI models exhibit scheming propensity, empirically confirming the dynamic classification problem.

⁴⁰Ley 25.326 de Proteccion de Datos Personales, Art. 5: data processing is unlawful without free, express, and informed consent of the data subject.

of commercial deployment. Data source: first documented cases of fraud exploiting multimodal agents, real-time voice cloning services, and autonomous embodied agents, tracked against their respective commercial launch dates. Falsification condition: a major new interaction technology reaches mass adoption without documented parasitic exploitation within 5 years.

Prediction 3 (niche vulnerability). Agentic fraud will disproportionately target populations exhibiting the niche conditions identified in Section V.B: high institutional trust, low digital literacy, weak social verification networks, and navigation of unfamiliar digital environments under stress. Data source: victim demographic profiles in fraud reports, 2026-2029, compared against demographic baselines. Falsification condition: agentic fraud victims show no demographic skew relative to the general population.

Prediction 4 (Herley-defense obsolescence). Consumer protection advice calibrated for r-strategy fraud ("if it sounds too good to be true"; "never give personal information to unsolicited callers") will show declining effectiveness as measured by the proportion of fraud victims who report having followed standard advice and still been defrauded. Data source: victim surveys in FTC and UFECI reports. Falsification condition: standard consumer advice remains equally effective against agentic and non-agentic fraud.

Prediction 5 (differential model colonization). Models with lower safety-training depth (as measured by single-prompt refusal rates) will be disproportionately used in agentic fraud pipelines relative to models with higher safety-training depth. Data source: forensic analysis of fraud tools identified by law enforcement, identifying which LLM APIs are used in detected fraud operations. Falsification condition: no correlation between single-prompt refusal rate and frequency of use in detected fraud tools.

Prediction 6 (orchestration-layer targeting). Regulatory interventions targeting only the model layer (guardrails, content filters) will show declining effectiveness over time as measured by the ratio of detected-to-attempted agentic fraud. Interventions targeting the orchestration layer (agentic framework restrictions, API use-pattern monitoring, adaptive planning detection) will show superior effectiveness. Data source: enforcement data from regulatory agencies, 2026-2030. Falsification condition: model-layer interventions prove as effective as or more effective than orchestration-layer interventions.

VIII. Discussion and Limitations

Five limitations deserve explicit acknowledgment.

First, the r-to-K framework, while heuristically useful, is a simplification. In evolutionary ecology, the r/K continuum has been substantially refined since MacArthur and Wilson's (1967) original formulation. Real organisms exhibit mixed strategies, and the continuum depends heavily on environmental conditions. The application to fraud strategies should be understood as an

analogy that captures the directionality of the shift (from low-investment mass-targeting to high-investment adaptive targeting) without claiming that fraud memes literally exhibit all features of K-selected biological organisms.

Second, Badhe's experimental design, while well-structured (270 experiments, three models, five scenarios, three user personas), uses simulated rather than real victims. The scripted user bot, though deliberately designed to introduce controlled resistance, cannot replicate the full range of human emotional responses, cultural variation, and contextual awareness that real scam targets bring to interactions. The dialogue completion rates Badhe reports should therefore be treated as upper bounds on real-world effectiveness for compliant personas and lower bounds for cautious ones. The true effectiveness of agentic fraud against real human targets remains an open empirical question that Badhe's framework is explicitly designed to facilitate answering through controlled experiments.

Third, the Argentine PAMI cases are documented through press reports, institutional communications, prosecutorial alerts, and lower court rulings rather than systematic data collection. A rigorous validation of the evolutionary predictions would require access to UFECI case files, PAMI's internal fraud databases, and longitudinal victim survey data, none of which is currently publicly available. The Argentine evidence should be understood as providing proof-of-concept grounding for the evolutionary framework, not as definitive empirical validation.

Fourth, the structural analogy between corporate Level 1 optimization and LLM Level 1 optimization is productive but not ontologically precise. Corporate optimization emerges from organizational routines, incentive structures, and aggregated human decisions. LLM optimization emerges from gradient descent on loss functions. Both lack genuine beliefs and moral reasoning, but the mechanisms through which they lack these features differ in ways that may prove legally relevant. Future work should formalize the conditions under which structural analogy suffices for liability analysis and the conditions under which mechanistic differences become legally decisive.

Fifth, the six falsifiable predictions in Section VII specify data sources and falsification conditions, but some depend on data that will not be available for 2-5 years. This is a limitation inherent to any predictive framework: the value of the predictions can only be assessed retrospectively. I include them precisely because unfalsifiable frameworks are scientifically unproductive.⁴¹ If the predictions fail, the framework should be revised.

Despite these limitations, the evolutionary framing offers analytical leverage that the cybersecurity framing does not. It explains why ScamAgent works (fraud memes are pre-adapted for social engineering; LLMs provide a new replication substrate). It predicts what happens next (any cost-reducing technology will be colonized; K-strategy demographics will shift). It identifies where regulation should focus (the agentic orchestration layer, not the model layer). And it grounds

⁴¹D.T. Campbell, "Evolutionary Epistemology," in P.A. Schilpp (ed.), *The Philosophy of Karl Popper* (Open Court, 1974). Campbell's "blind variation and selective retention" framework for the evolution of knowledge systems.

the analysis in documented real-world parasitism (Argentine PAMI campaigns) rather than purely hypothetical threat models.

IX. Conclusion

Badhe (2025) intended ScamAgent as a red-teaming exercise exposing gaps in LLM safety mechanisms. Read through the lens of Extended Phenotype Theory, Parasitic Spontaneous Order, and Asymmetric Intentionality Theory, and grounded in the Argentine PAMI fraud campaigns, it reveals something more consequential: the first empirically documented case of fraud memes colonizing artificial agents as reproductive machinery.

The shift from Herley-style filtering to adaptive persistence, from r-strategy to K-strategy parasitism, represents not an incremental increase in scam sophistication but a qualitative change in the evolutionary dynamics of deception. The extended phenotype of fraud no longer requires a human operator for each interaction. It requires only an agentic framework that can decompose malicious objectives into benign sub-tasks and pursue them across conversational turns with adaptive planning and persistent memory.

The Argentine case demonstrates that the memes ScamAgent automates are not laboratory constructs but field-tested parasites with decades of evolutionary refinement. PAMI scammers discovered, through trial and error, that institutional mimicry combined with urgency, decomposed data requests, and guided installation of remote access software could overcome the resistance of skeptical elderly victims. ScamAgent formalizes and scales this discovery, replacing the human scammer with a system that does not sleep, does not tire, does not feel guilt, and can conduct thousands of simultaneous conversations with individualized adaptive strategies.

The regulatory implication follows from the evolutionary analysis. Safety mechanisms designed for static, single-turn threats will fail against adaptive, multi-turn agents for the same reason that Herley-style consumer advice fails against K-strategy scammers: they assume a fixed adversary in a system that selects for adaptation. Effective regulation must target the agentic orchestration layer where adaptive capacity is concentrated, impose objective liability along the supply chain where risk is concentrated (following the Argentine jurisprudential trajectory from bank liability to infrastructure provider liability), accept that the adversary is not a fixed attacker but an evolving ecosystem of parasitic replicators, and anticipate that every reduction in the cost of adaptive interaction will be colonized.

Hayek observed that the extended order of human cooperation emerged without a designer. The extended order of human exploitation, it turns out, follows the same logic.

References

- Badhe, S. (2025). ScamAgents: How AI Agents Can Simulate Human-Level Scam Calls. *Proceedings of Machine Learning Research*, 299, 1-19.
- Campbell, D.T. (1974). Evolutionary Epistemology. In P.A. Schilpp (ed.), *The Philosophy of Karl Popper*. Open Court.
- Dawkins, R. (1976). *The Selfish Gene*. Oxford University Press.
- Dawkins, R. (1982). *The Extended Phenotype*. Oxford University Press.
- Dennett, D. (1987). *The Intentional Stance*. MIT Press.
- Federal Trade Commission. (2024). *Consumer Sentinel Network Data Book 2024*.
- Fiscales.gob.ar. (2025). Alertan por llamadas de WhatsApp fraudulentas que aparentan provenir de agentes del PAMI.
- Hayek, F.A. (1988). *The Fatal Conceit: The Errors of Socialism*. University of Chicago Press.
- Herley, C. (2012). Why Do Nigerian Scammers Say They Are from Nigeria? Microsoft Research.
- Hopman, K. et al. (2026). Scheming Reasoning Evaluations. arXiv:2603.01608v1.
- IJudicial.gob.ar. (2024). Condenan al banco Itau a indemnizar a un consumidor hipervulnerable.
- Infobae. (2025). Una auditoria en el PAMI detecto fraudes y falsificaciones en ordenes medicas.
- Infobae. (2026). Estafa informatica: la Justicia ordeno que un banco responda.
- JusticiaDePrimera.com. (2025). Se hacen pasar por empleados del PAMI y enganan a personas mayores.
- Lerer, I.A. (2026a). Law as a Primary Adaptive Platform. Zenodo. DOI: 10.5281/zenodo.18870552.
- Lerer, I.A. (2026b). Predatory Invitations as Extended Phenotype. Zenodo. DOI: 10.5281/zenodo.18853667.
- Lerer, I.A. (2026c). Intentionality Mismatch in Commercial Liability. Zenodo. DOI: 10.5281/zenodo.18903217.
- Lerer, I.A. (2026d). Spandrels of Accountability. Zenodo. DOI: 10.5281/zenodo.18882212.
- Lerer, I.A. (2026e). Synthetic Chaos as Institutional Laboratory. Zenodo. DOI: 10.5281/zenodo.18777434.
- MacArthur, R.H. and Wilson, E.O. (1967). *The Theory of Island Biogeography*. Princeton University Press.
- Ministerio de Seguridad de Mendoza. (2021). Seguridad advierte sobre posible estafa del cuento del tío.
- PAMI. (2025). Evitemos las estafas virtuales. <https://www.pami.org.ar/evitar-estafas>.
- Perfil. (2025). PAMI advierte sobre nuevas estafas digitales contra jubilados y pensionados.
- Sosis, R. and Alcorta, C. (2003). Signaling, solidarity, and the sacred. *Evolutionary Anthropology* 12(6): 264-274.
- STJ Corrientes. (2025). [Phishing; responsabilidad bancaria; discriminacion por edad]. Via Revista Quorum.

Zahavi, A. (1975). Mate selection: a selection for a handicap. *Journal of Theoretical Biology* 53(1): 205-214.

IA Use Disclaimer: I use Claude (Anthropic) as a research assistant for synthesis and drafting. The theoretical framework, the interpretations, the identification of connections between ScamAgent, EPT/PSO/AIT, and Argentine fraud cases, the falsifiable predictions, and all methodological decisions are my own. Any errors of fact or interpretation are mine.